Green University of Bangladesh

Department of Computer Science and Engineering (CSE)
Faculty of Sciences and Engineering
Semester: (Spring, Year: 2025), B.Sc. in CSE (Day)

Lab Report: 02

Course Title: Data Mining Lab
Course Code: CSE 436
Section: 212_D2

Lab Manual 03

Data Preprocessing Techniques

Student Details

Name	ID
Md. Almahmud	212002142

Submission Date : 30-04-2025
Course Teacher's Name : Md. Atikuzzaman

Lab Report Status

Marks : ……………………………………..Signature…………………………

Comments : …………………………………Date :…………………………….

1. TITLE OF THE LAB EXPERIMENT:

Data Preprocessing Techniques using Python

2. OBJECTIVES/AIM:

- To apply essential data preprocessing techniques on a dataset using Python.
- To handling missing values, duplicate data, inconsistent formatting, and preparing the dataset for analysis with visualizations.

3. MOTIVATION:

- Raw data often contains noise, missing values, or inconsistent formats, which can negatively affect the performance of machine learning models.
- Preprocessing ensures that the data is clean, reliable, and suitable for further analysis.

4. PROCEDURE:

- Load the “*Data_Cleaning_Dataset.csv*” dataset using Pandas.
- Inspect the dataset (shape, info, and head).
- Identify and handle missing values.
- Remove duplicates if any.
- Standardize column names and formats.
- Encode categorical data if needed.
- Save the cleaned dataset.

5. CODE:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler

file_path = '/content/Data_Cleaning_Dataset.csv'
df = pd.read_csv(file_path)

print(df.head())
print(df.info())
print("\nMissing Values Count:")
print(df.isnull().sum())
df_filled = df.copy()
for col in df_filled.columns:
    if df_filled[col].dtype == 'object':
        df_filled[col] = df_filled[col].fillna(df_filled[col].mode()[0])
    else:
        df_filled[col] = df_filled[col].fillna(df_filled[col].mean())
```

```
print("\nVisualizations Objets Type column: ")
categorical_cols = df_filled.select_dtypes(include='object').columns
for col in categorical_cols:
    plt.figure(figsize=(6, 4))
    sns.countplot(data=df_filled, x=col)
    plt.title(f'Count Plot of {col}')
    plt.xticks(rotation=45)
    plt.tight_layout()
    plt.show()

print("\nVisualizations Objets Type column: ")
numeric_cols = df_filled.select_dtypes(include=np.number).columns
for col in numeric_cols:
    plt.figure(figsize=(6, 4))
    sns.histplot(df_filled[col], kde=True, bins=30)
    plt.title(f'Distribution of {col}')
    plt.tight_layout()
    plt.show()

print("\nHeatmap of Correlation:")
plt.figure(figsize=(10, 6))
correlation = df_filled[numeric_cols].corr()
sns.heatmap(correlation, annot=True, cmap='coolwarm', fmt=".2f")
plt.title("Correlation Heatmap")
plt.tight_layout()
plt.show()

df_encoded = pd.get_dummies(df_filled)
scaler = StandardScaler()
scaled_data = scaler.fit_transform(df_encoded)
df_scaled = pd.DataFrame(scaled_data, columns=df_encoded.columns)
print("\n After Polish Scaled Data Sample:")
print(df_scaled.head())
```

6. OUTPUT:

```
print(df.head())
```

```
  EmployeeID  EmployeeName   Age  Department   JoiningDate     Salary
0    EID1001    Employee_1  19.0     Finance    2021-03-07    4901.91
1    EID1002    Employee_2  33.0     Finance    2023-10-09  200000.00
2    EID1003    Employee_3  65.0          HR    2022-02-07    4949.93
3    EID1004    Employee_4  65.0   Marketing    2022-04-25    7178.64
4    EID1005    Employee_5  23.0       Sales  "2020-07-08"    6673.17
```

```
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 160 entries, 0 to 159
Data columns (total 6 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   EmployeeID    160 non-null    object
 1   EmployeeName  160 non-null    object
 2   Age           146 non-null    float64
 3   Department    160 non-null    object
 4   JoiningDate   160 non-null    object
 5   Salary        145 non-null    float64
dtypes: float64(2), object(4)
memory usage: 7.6+ KB
None
```

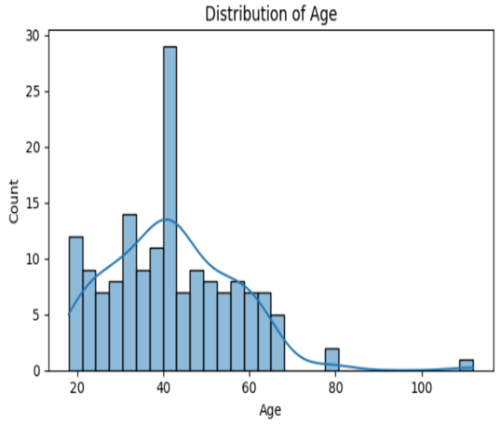
```
print("\nMissing Values Count:")
print(df.isnull().sum())
```

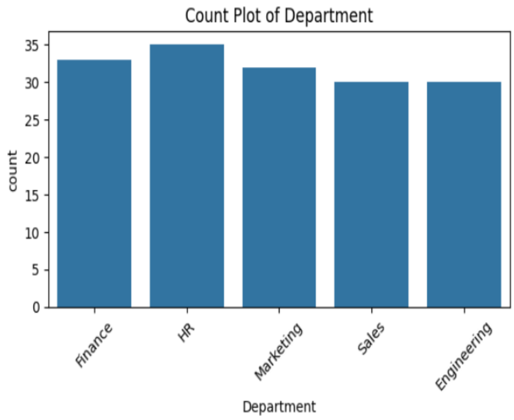
```
Missing Values Count:
EmployeeID       0
EmployeeName     0
Age             14
Department       0
JoiningDate      0
Salary          15
dtype: int64
```

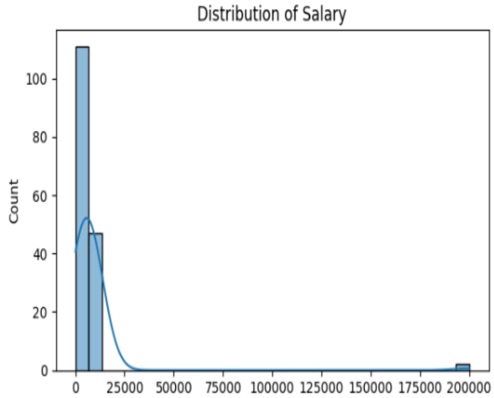
```
print("\n Check Missing Values Now:")
print(df_filled.isnull().sum())
```

```
 Check Missing Values Now:
EmployeeID      0
EmployeeName    0
Age             0
Department      0
JoiningDate     0
Salary          0
dtype: int64
```

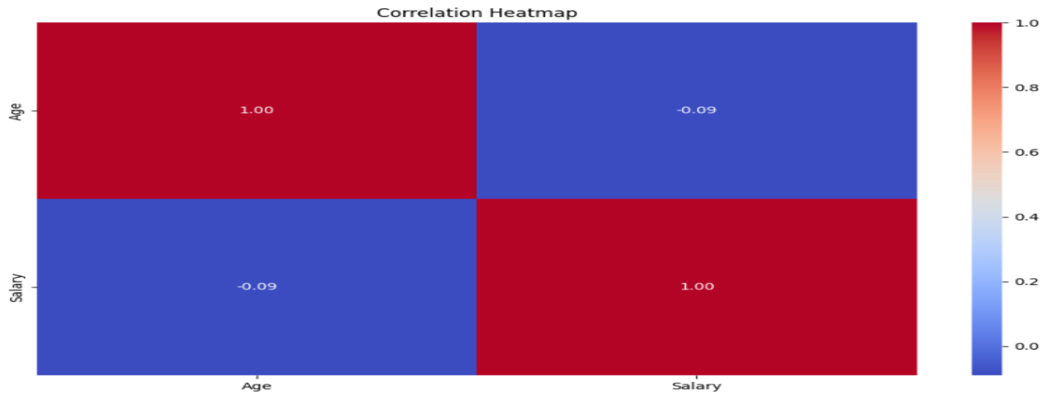
Visualizations Objets Type column:







Heatmap of Correlation:




```
After Polish Scaled Data Sample:
        Age    Salary  EmployeeID_EID1001  EmployeeID_EID1002  \
0 -1.566439 -0.151969           12.609520           -0.079305
1 -0.605802  8.861569           -0.079305           12.609520
2  1.589938 -0.149751           -0.079305           -0.079305
3  1.589938 -0.046784           -0.079305           -0.079305
4 -1.291971 -0.070137           -0.079305           -0.079305

   EmployeeID_EID1003  EmployeeID_EID1004  EmployeeID_EID1005  \
0           -0.112509           -0.112509           -0.079305
1           -0.112509           -0.112509           -0.079305
2            8.888194           -0.112509           -0.079305
3           -0.112509            8.888194           -0.079305
4           -0.112509           -0.112509           12.609520

   EmployeeID_EID1006  EmployeeID_EID1007  EmployeeID_EID1008  ...  \
0           -0.079305           -0.079305           -0.079305  ...
1           -0.079305           -0.079305           -0.079305  ...
2           -0.079305           -0.079305           -0.079305  ...
3           -0.079305           -0.079305           -0.079305  ...
4           -0.079305           -0.079305           -0.079305  ...

   JoiningDate_2024-07-23  JoiningDate_2024-08-10  JoiningDate_2024-08-14  \
0               -0.079305               -0.079305               -0.079305
1               -0.079305               -0.079305               -0.079305
2               -0.079305               -0.079305               -0.079305
```

7. DISCUSSION:

In the data cleaning and preprocessing phase, the dataset was first loaded and explored to understand its structure. This included checking column types, missing values, and overall data quality. Missing values were handled using forward fill, which replaces missing entries with the last known value. Depending on the dataset, other methods like mean imputation could also be considered. Duplicate rows were removed to maintain data integrity, and column names were standardized to make them easier to work with during analysis or modeling.

To get a better understanding of the data, some visualizations were used. A heatmap was created to clearly show where missing values existed. Histograms (histplots) helped visualize the distribution of numeric values like employee age and salary, showing how data points are spread. Countplots were useful for displaying the number of employees who joined during different time periods, giving insight into patterns.